

LEVENT CELIKSAN

Senior AI & Software Engineer | Technical Founder of Sealify

Agentic AI · Computer Vision · AI Systems · Backend

Istanbul, Türkiye | Open to remote work | Open to relocating to the U.S.

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SUMMARY

Senior AI & Software Engineer and technical founder with **19+ years** of experience building backend, infrastructure, and production software systems, including leading and mentoring engineers. Built and launched **Sealify** end to end as its sole technical founder, owning product architecture, backend engineering, applied AI, infrastructure, security, and operations.

Inventor on six U.S. provisional patent applications in AI-driven content protection, deepfake detection, and multimedia matching. Invited to participate in IPHatch North America 2026 with Sealify; the program is currently ongoing and final results are pending.

Specialties: custom computer vision algorithms on open-source models, agentic AI with local LLMs, and self-hosted AI infrastructure.

PROFESSIONAL EXPERIENCE

Technical Founder & Senior Principal AI Engineer — Sealify, Inc.

Nov 2023 – Present

Newark, Delaware (Remote) · Incorporated in Delaware in 2025

- **Product:** Launched Sealify in June 2026 as a multilingual (EN/TR/AR) SaaS platform with ~100 early users: Google sign-in, subscription plans (free to \$200/month) with crypto payments, a user dashboard, and an admin panel (users, credits, email, scheduled DMCA scans, user-approved takedown notices). Demo: youtu.be/LIoTl1nJU5c · Intro: youtu.be/Bh7kB-UIJhs
- **Measured results:** Validated large-scale web content scanning, video matching, and autonomous social media archiving through **3,000+ test runs**: scanned **200,000+** candidate images and identified **10,000+ verified matches across 5,000+ domains** at **~98% precision** in manual review of scan results; a single video test found **1,050+ matches across 80+ domains**; archiving tests automatically processed and registered **10,000+ media items (450+ GB)**.
- **Reverse image/video search:** Designed a two-stage reverse search: multi-engine retrieval (Google Lens, Google Reverse Image, Yandex, Bing) with automatic failover, then local verification. From a single photo, poster, or video frame, it finds public copies across the web in a **median of 2.5 minutes**, including resized, recolored, restyled, or audio-altered versions.
- **Matching engine:** Designed a custom multi-signal matching algorithm (color distribution, dominant palette, background, lighting, pHash, EfficientNet-B7 embeddings) gated by InsightFace/ArcFace face verification; calibrating its weights and thresholds raised matching accuracy from **~60% to ~97%**; runs on parallel workers in a queue-based, multi-instance architecture.
- **Self-hosted KYC:** Designed a multi-stage identity verification pipeline — custom template-based document recognition that accepts only passports, national IDs, and driver's licenses and rejects anything else, then a **blink-based liveness check** and face matching of the selfie against the document photo with open-source models — saving **~\$1 per verification** vs. paid providers such as Onfido or Jumio, with biometric data kept on Sealify's own infrastructure. Correctly verified **20/20 users** in a pilot test; documents with MRZ can be uploaded or photographed, while the selfie is captured live by camera only to reduce spoofing and replay attacks.
- **Novelty check & Bitcoin timestamping:** Before registration, videos pass an audio-first novelty check (audio recognition, then frame search), and content already online is rejected. Approved content is anchored to Bitcoin via OpenTimestamps as a fingerprint manifest (SHA-256 + pHash of 27 frames × 12 crop/rotation variants), with verifiable certificates and no transaction fees for users. **1,600+ registrations** anchored on mainnet, with independently verifiable timestamp proofs.
- **Autonomous monitoring:** Built worker systems that check users' connected Instagram, TikTok, and YouTube channels every 15 minutes, automatically fingerprinting new uploads and registering them on the blockchain; built a monitor that tracks piracy sites and feeds newly listed titles into the detection pipeline in near real time, plus a LangGraph agent with local LLMs that investigates suspected piracy sites end to end and writes evidence-based reports.
- **Deepfake detection:** Prototyped a hybrid detector for AI-generated images and videos, combining custom image-analysis signals (regional blur anomalies, homogeneous texture, noise, color and contrast distribution) with voting across Hugging Face classifiers, plus an asynchronous video analysis service that samples frames from submitted videos.
- **NCII & deepfake discovery:** Built a victim-protection feature on top of the reverse search and face-verification pipeline that finds potential non-consensual intimate and deepfake content; shipped in paid plans, it scans adult content platforms with privacy safeguards, processes hundreds of images per scan with Apple Silicon GPU acceleration, shows results blurred, and deletes data automatically.
- **Infrastructure, cost & business:** Optimized for long-term financial sustainability: planned RAM/CPU/GPU usage, server capacity, energy, and third-party service costs together; compared AWS, GCP, Azure, Vultr, and self-hosted options by workload and moved production to self-hosted hardware behind Cloudflare; managed Linux server security, backups, and monitoring. Bootstrapped and fully remote, with no office or paid LLM APIs: **~\$300/year** in recurring service costs; led financial planning and investor talks.

Senior Software Engineer — BUBU (Kopuzlar Group) · Türkiye

2018 – 2023

Set technical direction and led 2 developers; built Python APIs, e-commerce backends, and LLM/RAG integrations (2023); ran production infrastructure with Docker, CI/CD, monitoring, and backups.

- Software Engineer — Artworks Lab AB** · Stockholm, Sweden (Remote) 2015 – 2018
Boutique e-commerce agency: led backend development and built Django/DRF e-commerce systems, using Elasticsearch, Redis, and AWS to support search, caching, and high-traffic campaigns.
- Software Engineer — Yellow Pages / DataWorks** · United States (Remote) 2012 – 2015
U.S. local search and digital marketing company: developed business-listing, mapping/geocoding, and caller-ID services, including REST APIs and backend systems using PHP/Laravel, MySQL, and AWS.
- Software Engineer — Jolly Tur** · Istanbul, Türkiye 2008 – 2011
Major Turkish travel company: developed PHP/MySQL booking and e-commerce backends, SOAP/XML supplier integrations, and supported Linux/Windows server infrastructure.
- Junior Software Engineer — REKLAMARKA** · Türkiye 2006 – 2008
Digital agency: developed corporate websites, PHP/MySQL backends, admin panels, and e-commerce integrations for client projects.

PATENT APPLICATIONS & WRITING

Six U.S. provisional patent applications (USPTO, Nov–Dec 2025, Patent Pending), including content tracking and protection (63/923,341); real-time deepfake detection (63/923,358); multimodal novelty assessment (63/923,342); autonomous video registration (63/934,369); multimedia semantic equivalence (63/946,066).

Article: “How I find edited copies of images across the web without crawling it” (Sep 2026) · leventceliksan.github.io/copy-detection

SELECTED AI PROJECTS

Aesthetic Detector — BEAUTY (Android) 2025 – Present

Personal product: Android app built in Kotlin with a credit-based revenue model; preparing for Google Play launch

- Fine-tuned a CLIP ViT-B/16-based model on **1,000+ portrait photos** to classify signs of cosmetic intervention, then converted it to ONNX and optimized it to run inference entirely on-device, returning a result in **~2 seconds**.
- Developed a multi-stage scoring pipeline combining the overall classification score with EfficientNet-B0-based regional facial analysis; Google ML Kit for face detection and pose checks, ONNX Runtime for inference.
- Privacy-oriented architecture: heavy AI processing runs on the user's device, so images never need to be sent to a server; the Node.js, Express, and MySQL backend handles only authentication, the credit system, data synchronization, and coordination.

AI Advertising Agency Automation 2025 – Present

Personal project

- Built a generative AI advertising agency with **7 agents** (copywriter, photographer, designer, social media, web, video, assistant) coordinated by a manager agent that turns natural-language chat requests into role-specific briefs on a shared SQLite task board, producing social posts, banners, campaign sites, and Reels.
- Built a single LLM gateway in Python using FastAPI and Pydantic that picks Gemini model tiers by task importance, falls back automatically to local Ollama models on quota limits or errors, supports OpenRouter models, and tracks tokens and cost per job; designs are rendered from HTML/CSS with Playwright and videos encoded with FFmpeg locally, keeping server costs near zero.
- Developed autonomous quality control: deterministic checks (missing files, banned words, font sizes measured before rendering), vision-model review of outputs at real mobile size for legibility and overflow, automatic prompt correction with up to 3 image regenerations, rotation across 12 layouts to prevent repetitive designs, and revisions that rerun only the requested agent.

TECHNICAL SKILLS

- **Generative AI & agents:** LLMs, agentic AI, multi-agent systems, LangGraph, LangChain, tool/function calling, structured outputs, LLM evaluation, guardrails, multi-LLM routing and failover, Ollama, Gemini API, OpenRouter, model fine-tuning
- **RAG & retrieval:** embeddings, semantic search, vector search (FAISS, pgvector, Qdrant), hybrid retrieval, reranking
- **ML & computer vision:** PyTorch, Hugging Face Transformers, ONNX / ONNX Runtime, on-device inference, OpenCV, InsightFace/ArcFace, face matching, liveness detection, EfficientNet, CLIP, DINOv2, YOLOv8, NumPy, Pandas, scikit-learn, perceptual hashing, deepfake detection, ID document processing (MRZ), OCR/Tesseract, Whisper
- **Backend & data:** Python (FastAPI, AsyncIO, Pydantic, Flask, Django/DRF, SQLAlchemy, Celery, pytest), Node.js/Express, REST, WebSockets, JWT; MySQL, PostgreSQL, MongoDB, SQLite, Redis, Elasticsearch, RabbitMQ
- **Mobile, automation & blockchain:** Android (Kotlin, Google Play Billing, ML Kit); Playwright, FFmpeg; OpenTimestamps / Bitcoin anchoring, SHA-256
- **Cloud & DevOps:** AWS, GCP, Azure, Vultr, self-hosted; Linux, Nginx, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, GitLab CI, Jenkins), Prometheus, Grafana, Sentry, Cloudflare; capacity planning and cost modeling
- **Systems & architecture:** Model Context Protocol (MCP), AI observability and tracing, microservices, event-driven architecture, distributed systems, OAuth2/RBAC

EDUCATION & LANGUAGES

Collège LaSalle, Montréal, Canada — Bachelor's Degree, Computer Science Technology 2002 – 2006

Languages: Turkish — Native | English — Professional working proficiency